

Project Topics

1. Differential Privacy and Online Learning

Stability is central to many regret analyses: the distribution of an algorithm’s next decision should not change too much after one additional loss. This paper uses differential privacy to formalize stability for randomized online algorithms and turn it into regret bounds. The resulting framework applies to online optimization and prediction with expert advice.

- Jacob Abernethy, Young Hun Jung, Chansoo Lee, Audra McMillan, and Ambuj Tewari. *Online Learning via the Differential Privacy Lens* (NeurIPS 2019).
- Gonen, Alon and Hazan, Elad and Moran, Shay. *Private learning implies online learning: An efficient reduction* (NeurIPS 2019).

2. Fair Classification via No-Regret Learning

Fair classification adds constraints on how a classifier performs across groups. Minimizing prediction error subject to these constraints can be formulated as a game between a learner choosing classifiers and an auditor identifying violated constraints. No-regret dynamics yield a mixture of classifiers with bounds on both error and constraint violations. The subgroup-fairness paper addresses the difficulty of auditing many overlapping groups.

- Alekh Agarwal, Alina Beygelzimer, Miroslav Dudík, John Langford, and Hanna Wallach. *A Reductions Approach to Fair Classification* (ICML 2018).
- Michael Kearns, Seth Neel, Aaron Roth, and Zhiwei Steven Wu. *Preventing Fairness Gerrymandering: Auditing and Learning for Subgroup Fairness* (ICML 2018).

3. Human–AI Collaboration and Agreement

A human expert and a model may make different predictions because they have access to different information. These papers study protocols in which the two repeatedly exchange predictions and revise them. Calibration provides a condition under which disagreement can be used to improve prediction accuracy. The online formulation connects these protocols to no-regret learning.

- Natalie Collina, Surbhi Goel, Varun Gupta, and Aaron Roth. *Tractable Agreement Protocols* (STOC 2025).
- Natalie Collina et al. *Collaborative Prediction: Tractable Information Aggregation via Agreement* (SODA 2026).

4. AI Alignment: Preference Optimization and Welfare

Preference optimization trains a model using pairwise comparisons between responses. The Nash-learning formulation treats these comparisons as a game and seeks an equilibrium using no-regret learning. The main paper asks how well preference-based objectives maximize average user utility when users value responses differently. Pairwise comparisons reveal users’ rankings; welfare also depends on how strongly they prefer one response to another.

- Paul Gözl, Nika Haghtalab, and Kunhe Yang. *Distortion of AI Alignment: Does Preference Optimization Optimize for Preferences?* (NeurIPS 2025; main paper).
- Rémi Munos et al. *Nash Learning from Human Feedback* (ICML 2024; background on the objective).

5. Multi-Distribution Learning

In multi-distribution learning, a single predictor must perform well on each of several data distributions. For example, a model trained on data from several hospitals should have low error at every hospital. The objective is to minimize the worst expected loss. A zero-sum formulation pairs a learner choosing predictors with an adversary choosing weights over distributions. These papers combine no-regret learning with adaptive sampling to reduce the total sample complexity.

- Nika Haghtalab, Michael I. Jordan, and Eric Zhao. *On-Demand Sampling: Learning Optimally from Multiple Distributions* (NeurIPS 2022).
- Zihan Zhang, Wenhao Zhan, Yuxin Chen, Simon S. Du, and Jason D. Lee. *Optimal Multi-Distribution Learning* (COLT 2024).

6. Incentive Design for Learning Agents

This project studies how a mediator can use payments to steer no-regret learners toward a desired equilibrium. The players continue to run their own learning algorithms, so the mediator must account for their responses over time. The questions are which equilibria can be reached this way and how much the mediator must pay, including whether the average payment per round can vanish.

- Brian Hu Zhang et al. *Steering No-Regret Learners to a Desired Equilibrium* (EC 2024; main paper).
- Guru Guruganesh et al. *Contracting with a Learning Agent* (NeurIPS 2024; related reading).

7. Online Conformal Prediction

Online conformal prediction constructs prediction intervals for data that may change over time. Past coverage errors are used to update the intervals, with the goal of achieving a target coverage frequency such as 90%. The challenge is to maintain coverage with reasonably narrow intervals as the distribution changes. The update rules connect this problem to online optimization and step-size selection.

- Isaac Gibbs and Emmanuel Candès. *Adaptive Conformal Inference Under Distribution Shift* (NeurIPS 2021).
- Isaac Gibbs and Emmanuel J. Candès. *Conformal Inference for Online Prediction with Arbitrary Distribution Shifts* (Journal of Machine Learning Research, 2024).

8. Counterfactual Regret Minimization (CFR)

A pure strategy in an extensive-form game specifies an action at every information set. Running a regret minimizer over all such strategies is usually impractical. CFR instead minimizes counterfactual regret separately at each of a player's information sets. A regret decomposition then bounds overall regret in terms of these local regrets. In two-player zero-sum games with perfect recall, this yields convergence of the average strategies to Nash equilibrium.

- Martin Zinkevich, Michael Johanson, Michael Bowling, and Carmelo Piccione. *Regret Minimization in Games with Incomplete Information* (NIPS 2007).
- Marc Lanctot, Kevin Waugh, Martin Zinkevich, and Michael Bowling. *Monte Carlo Sampling for Regret Minimization in Extensive Games* (NIPS 2009).

9. Omniprediction

An omnipredictor supports decisions under many loss functions. Once trained, its output is post-processed for the chosen loss, and the resulting decisions should compete with the best predictor in a specified class for that loss. The main paper obtains such guarantees for convex losses using multicalibration, which requires calibration relative to a class of test functions. The second paper studies an online formulation and its connection to no-regret learning.

- Parikshit Gopalan, Adam Tauman Kalai, Omer Reingold, Vatsal Sharan, and Udi Wieder. *Omnipredictors* (ITCS 2022; especially Sections 5–6).
- Sumegha Garg, Christopher Jung, Omer Reingold, and Aaron Roth. *Oracle Efficient Online Multicalibration and Omniprediction* (SODA 2024).